



**Generation Interconnection
System Impact Study Report
for
Queue Project AF1-337
LAURELTON-MIFFLINBURG 69 KV
12 MW Capacity / 20 MW Energy**

August 2020

Table of Contents

1	Introduction.....	3
2	Preface.....	3
3	General	3
4	Point of Interconnection.....	4
5	Cost Summary	4
6	Transmission Owner Scope of Work	5
7	Attachment Facilities	6
8	Direct Connection Cost Estimate.....	6
9	Non-Direct Connection Cost Estimate.....	6
10	Schedule.....	7
11	Interconnection Customer Requirements.....	7
	11.1 PPL EU Interconnection Requirements.....	7
	11.2 IC Direct Transfer Trip (DTT) Requirements	7
12	Revenue Metering and SCADA Requirements	8
	12.1 PJM Requirements	8
	12.2 Meteorological Data Reporting Requirements	8
	12.3 Interconnected Transmission Owner Requirements.....	8
13	Summer Peak Analysis	9
	13.1 Generation Deliverability	9
	13.2 Multiple Facility Contingency	9
	13.3 Contribution to Previously Identified Overloads.....	9
	13.4 Steady-State Voltage Requirements	9
	13.5 Potential Congestion due to Local Energy Deliverability.....	9
14	Light Load Analysis	9
15	Short Circuit Analysis.....	10
16	Stability and Reactive Power	10
17	Affected Systems	10
	17.1 NYISO	10
18	Attachment 1: One Line Diagram	11
19	Attachment 2: Project Location	12

1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is PPL.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), **PD Solar 5 LLC**, has proposed a Solar generating facility located in Union County, Pennsylvania. The installed facilities will have a total capability of 20 MW with 12 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is March 01, 2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-337
Project Name	LAURELTON-MIFFLINBURG 69 KV
State	Pennsylvania
County	Union
Transmission Owner	PPL
MFO	20
MWE	20
MWC	12
Fuel	Solar
Basecase Study Year	2023

New Service Customers proposing queue projects that can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF1-337 will interconnect with the PPL transmission system tapping the Sunbury to Lock Haven 69 KV line between the **Laurelton Tap and Mifflinburg Tap 69 kV** buses. The Point of Interconnection (POI) will be at the PPL EU owned termination structure where the Interconnection Customer's transmission line terminates (with insulators).

5 Cost Summary

The AF1-337 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ 0
Allocation towards System Network Upgrade Costs*	\$ 0
Total Costs	\$ 0

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

6 Transmission Owner Scope of Work

Note: The cost for the physical interconnection will be borne by the AF1-333 project. The information below is provided for reference only. If AF1-333 withdraws from the PJM queue, this project could become responsible for the physical interconnection costs.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Attachment Facilities	\$ 723,600
Direct Connection Network Upgrade	\$ 0
Non Direct Connection Network Upgrades	\$ 556,400
Total Physical Interconnection Costs	\$ 1,280,000

PPL EU can accommodate this interconnection by constructing a new transmission line tap off the Sunbury-Lock Haven 69kV line and completing associated remote end relay work.

Risks and Assumptions

- No major environmental, real estate, siting, or permitting issues
- IC is responsible for acquisition of easements, permits, and right of way for any Direct Connection Network Upgrades and Attachment Facilities per PPL EU standards and requirements.
- PPL EU will perform all grading, site preparation, and establish access roads for the PPL EU owned Attachment Facilities per PPL EU standards and requirements.

7 Attachment Facilities

69 kV Transmission Line Tap

The Attachment Facilities will connect to the Sunbury-Lock Haven 69kV line approximately 2.5 miles from the Laurelton Tap and 0.16 miles from the Mifflinburg Tap. This scope of work is based on transmission tap GPS Coordinates: 40°54'8.44"N, 77° 3'50.66"W.

PPL EU work will consist of the following:

- Replace existing H-frame structure at PPL Grid 18942N26880 with a new tap structure on the Sunbury-Lock Haven 69 kV line
- Install a motor operated switch and structure
- Install a terminal structure (POI structure)
- Install ~400' of wire utilizing 556 ACSR from the tap structure to the POI structure

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
69kV Transmission Tap	\$ 723,600
Total Attachment Facility Costs	\$ 723,600

8 Direct Connection Cost Estimate

None

9 Non-Direct Connection Cost Estimate

Remote End Work – Sunbury 69kV Substation

- Complete remote end relay work at Sunbury 69 kV Substation for Direct Transfer Trip.
- Model IC in CAPE and conduct a wide area short-circuit study two busses away from the IC facilities. Identify affected relays and revise settings as needed.
- Conduct a review of the IC relay settings and engineering package (submitted by IC to PPL EU).

Remote End Work – Lock Haven 69 kV Substation

- Complete remote end relay work at Lock Haven 69 kV Substation for Direct Transfer Trip.
- Model IC in CAPE and conduct a wide area short-circuit study two busses away from the IC facilities. Identify affected relays and revise settings as needed.

- Conduct a review of the IC relay settings and engineering package (submitted by IC to PPL EU).

Sunbury-Lock Haven 69 kV Line Modifications to tie in the AF1-333 Attachment Facilities

- Tie the new AF1-333 Attachment Facilities into the Sunbury- Lock Haven 69 kV line
- Reframe structures adjacent to the new tap structure to tension

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Remote End Work at Sunbury Substation	\$ 238,000
Remote End Work at Lock Haven Substation	\$ 238,000
Sunbury-Lock Haven 69 kV Line Modifications to tie in the AF1-333 Attachment Facilities	\$ 80,400
Total Non-Direct Connection Facility Costs	\$ 556,400

10 Schedule

The estimated time to complete the scope of work is 12-18 months after the PJM three-party Interconnection Service Agreement (ISA) and Interconnection Construction Service Agreement (ICSA) are signed and PPL EU receives Notice to Proceed from the IC.

11 Interconnection Customer Requirements

11.1 PPL EU Interconnection Requirements

PPL EU applicable technical standards that address requirements for interconnection of generation, transmission, and end user facilities can be found at the following link:

<https://pjm.com/planning/design-engineering/to-tech-standards/private-ppl.aspx>

11.2 IC Direct Transfer Trip (DTT) Requirements

PPL EU will require an independent communication path, for Direct Transfer Trip (DTT) of the IC Intertie Protective Relaying (IPR) Fault Interrupting Devices (FIDs), consisting of one communication circuit with the Sunbury 69kV Substation and one communication circuit with the Lock-Haven 69kV substation.

PPL EU does not have OPGW available on the Sunbury-Lock Haven 69 kV line available for DTT to the Sunbury and Lock Haven 69kV Substations. PPL EU assumes that the IC will procure the independent communication path through a third-party provider. Upon request, PPL EU will evaluate the feasibility of installing OPGW the Sunbury-Lock Haven 69 kV line for DTT.

To ensure reliable communication, the IC shall also provide DTT relaying equipment identical to the PPL EU DTT relaying equipment. All DTT relaying equipment shall connect to the respective communication path. All DTT relaying equipment should reside within the same location as the IPR and Point of Contact (POC) relaying equipment.

12 Revenue Metering and SCADA Requirements

12.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

12.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

12.3 Interconnected Transmission Owner Requirements

Installation of revenue grade Bi-directional Metering Equipment will be required in the vicinity of the POI to measure kWh and kVARh. PPL EU will design and supply the required metering equipment; all installation costs would be borne by the IC including CTs/PTs. All metering equipment must meet applicable PPL EU tariff requirements as well as being compliant with all applicable requirements of the PJM agreements. The equipment must provide bidirectional revenue metering (kWh and kVARh) and real-time data (kW, kVAR, circuit breaker status, and generator bus voltages) for the IC's generating resource. The metering equipment should be housed in a control cabinet or similar enclosure and must be accessible to PPL EU metering personnel.

13 Summer Peak Analysis

The Queue Project AF1-337 was evaluated as a 20.0 MW (Capacity 12.0 MW) injection tapping the **Laurelton TP - Mifflinburg TP 69 kV** line in the PPL area. Project AF1-337 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-337 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

13.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

13.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

13.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

13.4 Steady-State Voltage Requirements

None

13.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

14 Light Load Analysis

Light Load Analysis is not required.

15 Short Circuit Analysis

The following Breakers are overdutied:

None

16 Stability and Reactive Power

(Summary of the VAR requirements based upon the results of the dynamic studies)

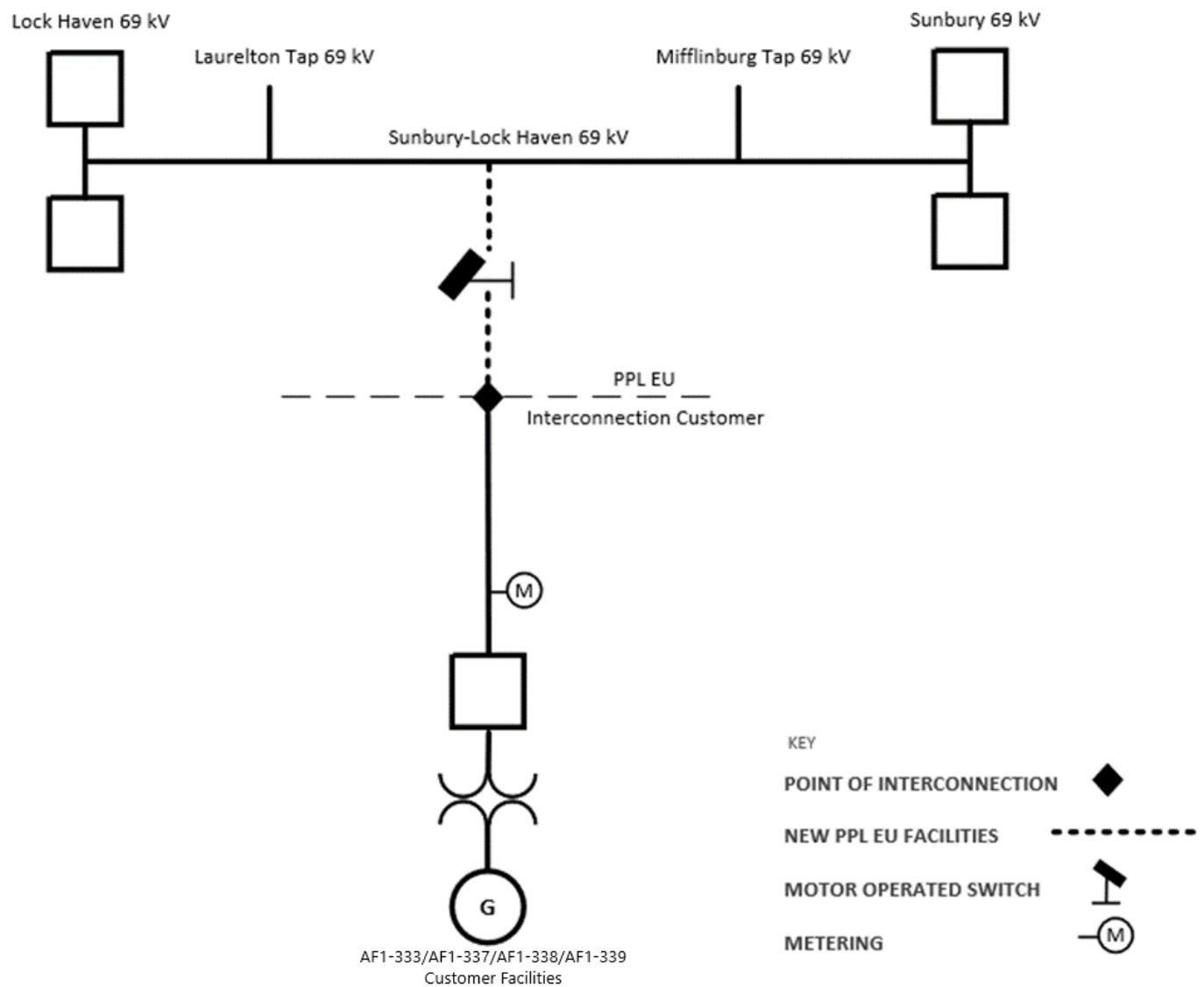
To be determined in the Facilities Study Phase. Additional system reinforcements may be identified to mitigate any stability concerns.

17 Affected Systems

17.1 NYISO

NYISO Impacts to be determined during later study phases (as applicable).

18 Attachment 1: One Line Diagram



19 Attachment 2: Project Location

